

IPv4 addressing and routing

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*Alessio Sacco, Claudio Casetti, Guido Marchetto, Marco
Rapelli*



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The IP protocol: a recap

- *IP (Internet Protocol)*: the network layer protocol of the TCP/IP architecture (RFC 791)
 - Defines the datagram format, the addressing plan and the forwarding rules
- *Connectionless*: no connection setup, each datagram is forwarded independently of the others
 - Datagrams of the same flow can follow different paths and arrive out of order
- *Best effort*: the network does its best to deliver datagrams, but guarantees nothing
 - Datagrams can be lost, duplicated, delayed, corrupted
 - No guarantee on throughput or delay, no acknowledgements, no retransmissions
- Error recovery, when needed, is left to the *end systems* (e.g., TCP)
 - Routers keep no per-flow state: simple core, intelligence at the edges



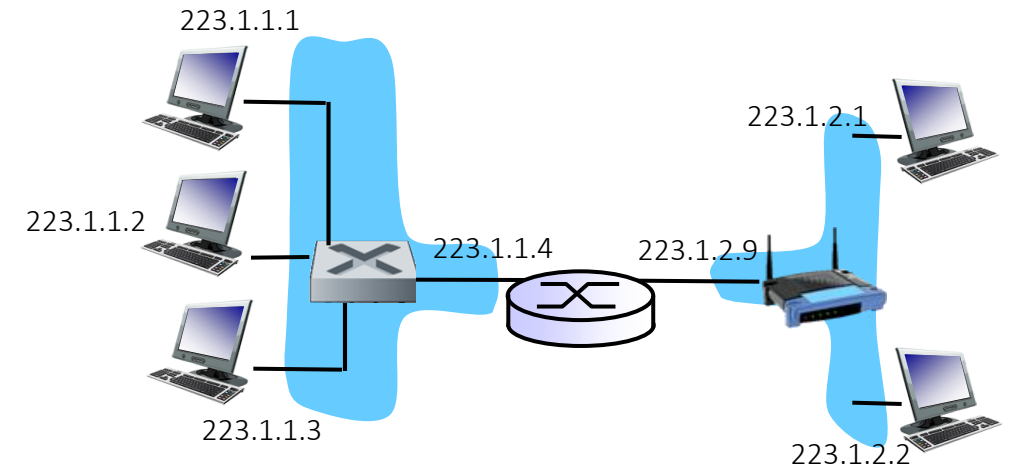
The IP protocol: a recap

- *Global addressing*: 32-bit addresses identifying interfaces
 - Hierarchical structure: network part + host part
- *Hop-by-hop forwarding*: each router decides on its own
 - Based on the destination address only, looking up its routing table
- *Independent of the underlying technology*
 - IP runs over any link layer: Ethernet, Wi-Fi, PPP, ...
 - *Fragmentation* adapts the datagram to the MTU of each link
- *Loop protection*: TTL decremented at every hop, dropped at zero
- *Header checksum*: protects header, not payload
- *Companion protocols*: ICMP and ARP
 - ICMP for error reporting and diagnostics, ARP for address resolution



IP addressing: terminology

- *IP address*: 32-bit identifier for host, router interfaces
- *Network part*: high order bits of the IP address
- *Host part*: low order bits of the IP address
- *IP network*: set of IP devices whose interfaces
 - Have the same network part of the IP address
 - Are connected to the same physical (link-layer) network



IP addressing: special addresses

Some value	All 0s
All 1s	
Some value	All 1s
127	Anything (often 1)

the (sub)network ID

limited broadcast (local net)

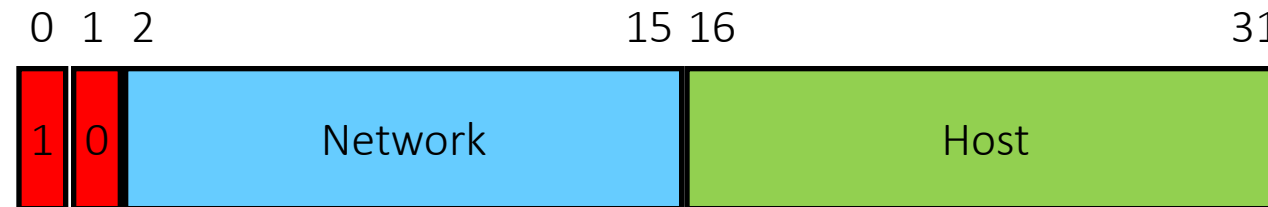
directed broadcast for net

loopback

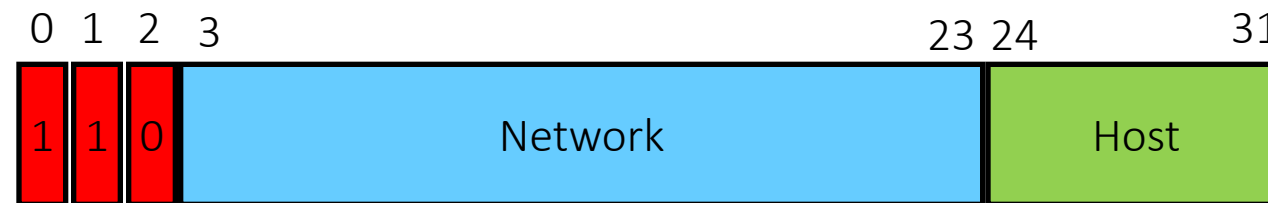
IP addressing classes



Class A – 128 networks – 1^o byte: 0-127



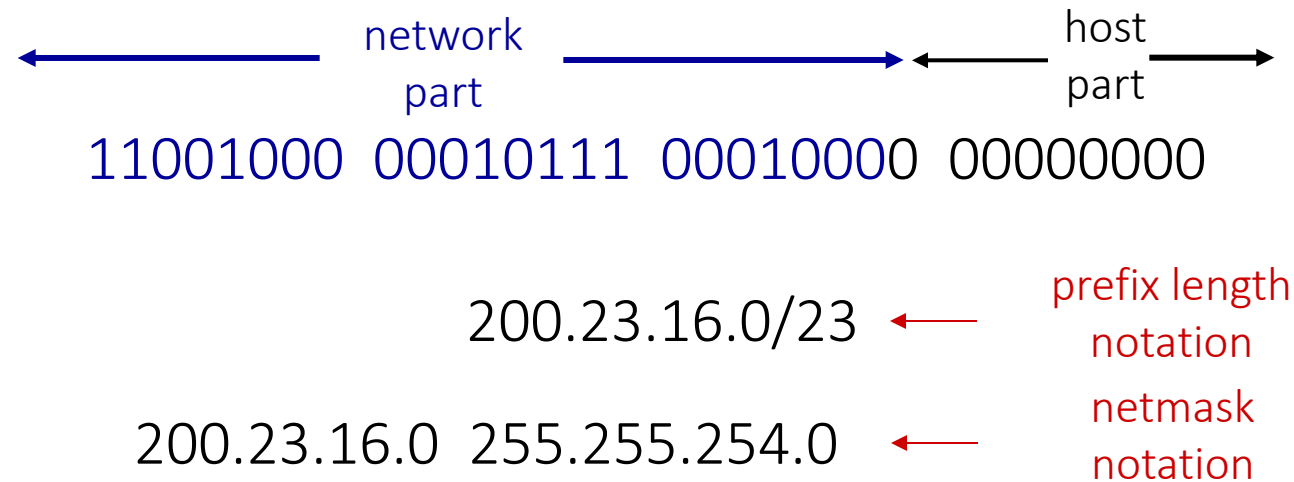
Class B – 16K networks – 1^o byte: 128-191



Class C – 2M networks - 1^o byte: 192-223

IP addressing: CIDR

- **CIDR**: **C**lassless **I**nter**D**omain **R**outing
 - network portion of address of arbitrary length
 - address format: *network ID* + *prefix length* or *netmask*
 - *prefix length*: /x, where x is # bits in network portion of address
 - *netmask*: all '1s' in the network part, all '0s' in the host part



IP addressing: CIDR

- *Valid netmasks*: possible values in the 4 bytes composing the address

0	0000	0000	(256)	
128	1000	0000	(128)	
192	1100	0000	(64)	
224	1110	0000	(32)	
240	1111	0000	(16)	
248	1111	1000	(8)	
252	1111	1100	(4)	smaller usable netmask
254	1111	1110	(2)	not valid in the 4° byte
255	1111	1111	(1)	represents the single device

IP addressing: CIDR

■ *Some examples*

- 130.192.0.0/16 – 130.192.0.0 255.255.0.0
- 130.192.0.0/24 – 130.192.0.0 255.255.255.0
- 130.192.0.0/25 – 130.192.0.0 255.255.255.128
- 130.192.2.0/23 – 130.192.2.0 255.255.254.0
- 130.192.1.4/30 – 130.192.1.4 255.255.255.252
- ~~130.192.1.0/31 – 130.192.1.0 255.255.255.254~~

Each IP network *must* contain at least the network ID and the broadcast address!



IP addressing: CIDR

■ Valid Network ID

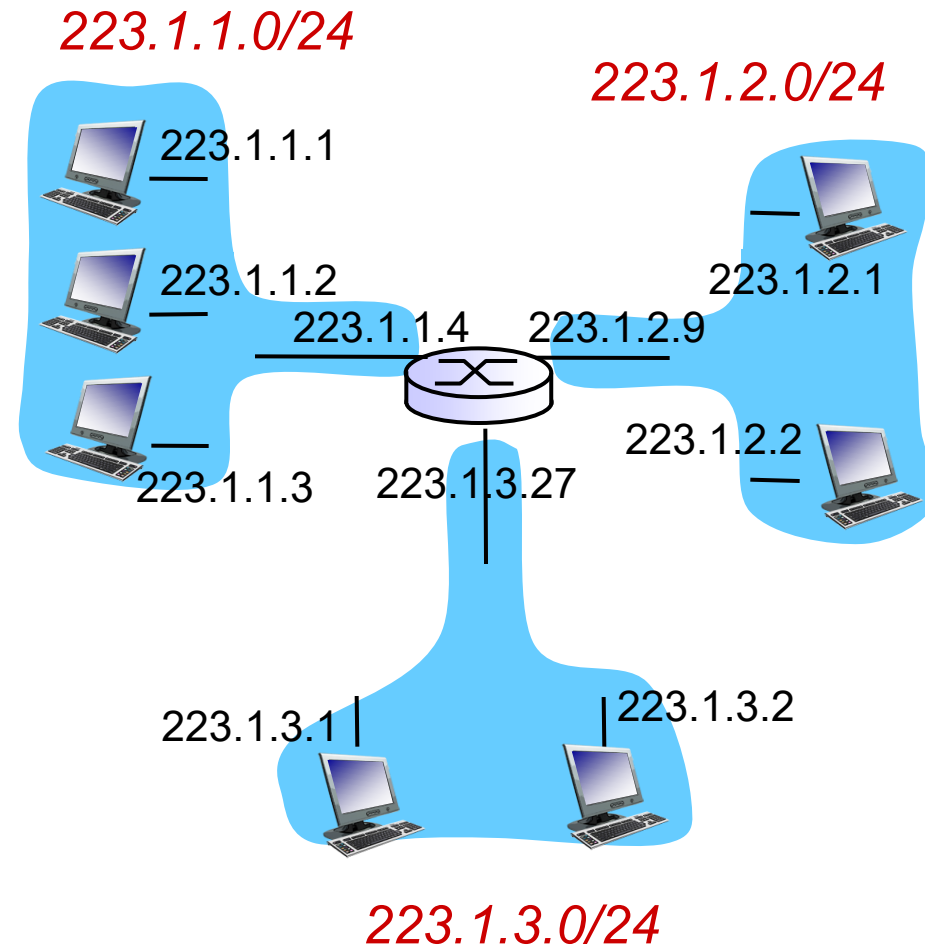
- 130.192.1.4/30
- 130.192.1.16/30
- 130.192.1.16/29

■ Invalid Network ID

- 130.192.1.1/30
- 130.192.1.4/29
- 130.192.1.24/28



IP addressing: a real example



netmask: 255.255.255.0

IP routing

■ *The general rules:*

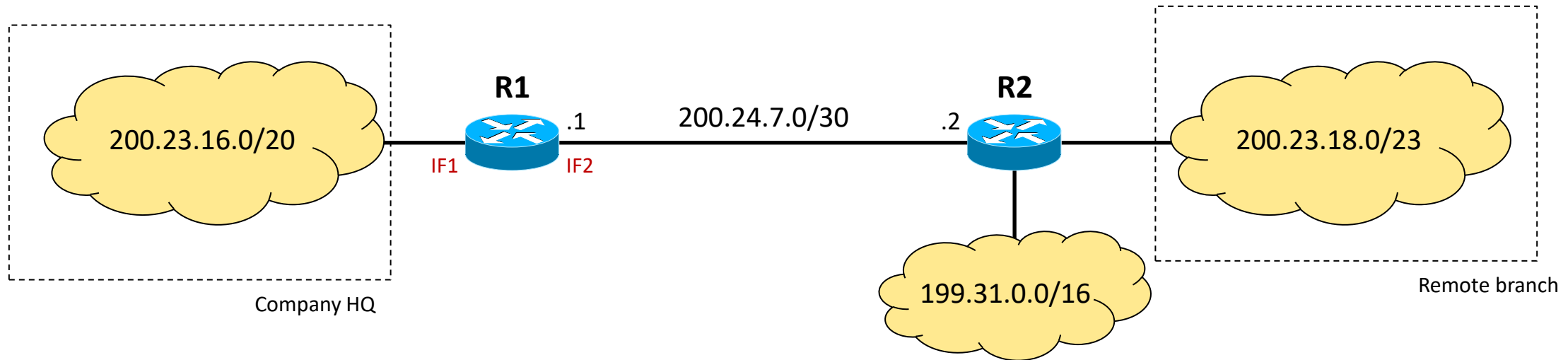
- Given a destination IP address to reach, an IP device searches its own routing table, looking for a *match*
- In case of multiple matches, it selects the most specific one (*longest prefix matching*)

routing table	
destination	output link
200.23.16.0/20	1
200.23.18.0/23	2
199.31.0.0/16	2



A real Routing Table

- A complete routing table also says to whom the packet is handed on each link



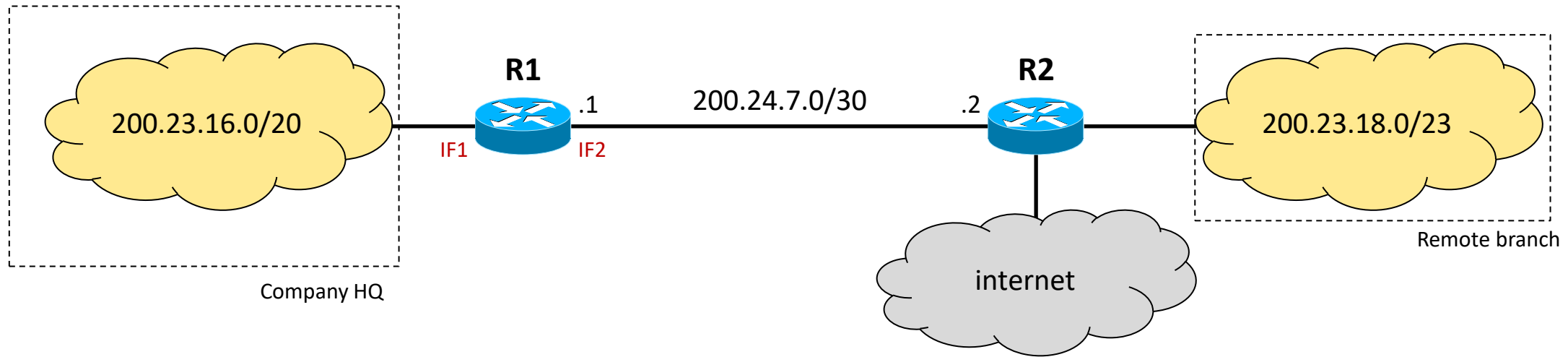
Routing table of R1		
Destination	Output link	Next hop
200.23.16.0/20	IF1	direct
200.23.18.0/23	IF2	200.24.7.2
199.31.0.0/16	IF2	200.24.7.2
200.24.7.0/30	IF2	direct

Longest prefix matching: a packet for 200.23.19.7 matches both 200.23.16.0/20 and 200.23.18.0/23. The /23 entry is more specific, so R1 sends it on link 2 to the next hop 200.24.7.2.

Direct routes: networks attached to R1, including the /30 point-to-point link, need no next hop; the packet is delivered on the link itself.

A real Routing Table: what about other routes?

- Destinations beyond R2 can be collapsed into a single default entry



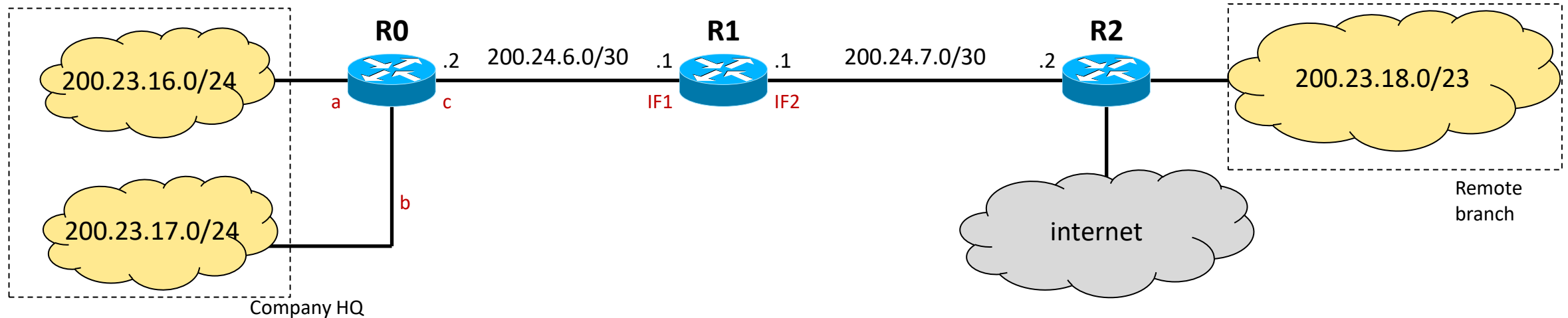
Routing table of R1		
Destination	Output link	Next hop
200.23.16.0/20	IF1	direct
200.23.18.0/23	IF2	200.24.7.2
200.24.7.0/30	IF2	direct
0.0.0.0/0 (default)	IF2	200.24.7.2

Default route: 0.0.0.0/0 matches every destination with the shortest possible prefix, so it is used only when no other entry matches.

The /23 entry must stay: without it, packets for 200.23.18.0/23 would match 200.23.16.0/20 and leave on link 1.

A real Routing Table: the Direct Delivery problem

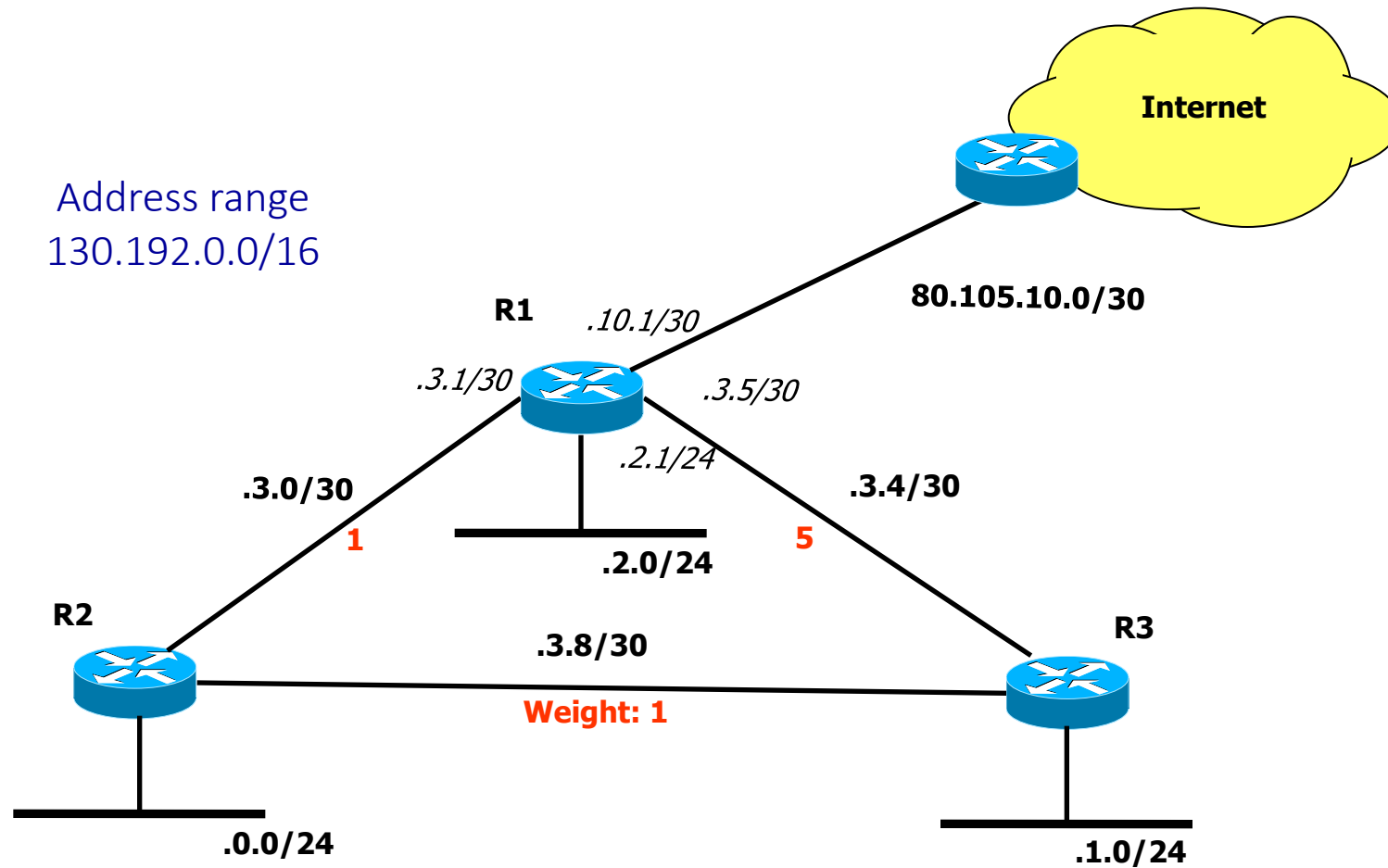
- ISP A holds 200.23.16.0/20 and routes it internally; 200.23.18.0/23 is now behind R2



Routing table of R0		
Destination	Output link	Next hop
200.23.16.0/24	a	direct
200.23.17.0/24	b	direct
200.24.6.0/30	c	direct
200.23.18.0/23	c	200.24.6.1
0.0.0.0/0 (default)	c	200.24.6.1

Routing table of R1		
Destination	Output link	Next hop
200.23.16.0/20	1	200.24.6.2
200.23.18.0/23	2	200.24.7.2
200.24.6.0/30	1	direct
200.24.7.0/30	2	direct
0.0.0.0/0 (default)	2	200.24.7.2

IP routing: an example



IP routing: an example

R1 routing table

<i>Type</i>	<i>Destination</i>	<i>Next-hop</i>	<i>Cost</i>
S	130.192.0.0/24	130.192.3.2	1
S	130.192.1.0/24	130.192.3.2	2
S	130.192.3.8/30	130.192.3.2	1
S	0.0.0.0/0	80.105.10.2	1
C	130.192.2.0/24	130.192.2.1	0
C	130.192.3.0/30	130.192.3.1	0
C	130.192.3.4/30	130.192.3.5	0
C	80.105.10.0/30	80.105.10.1	0

remote
interface!!

local
interface!!

